Probabilistic Artificial Intelligence
Questions Pack

January 19, 2026

Time limit: 120 minutes

Instructions. This pack contains all questions for the final exam. It contains the questions only. You must provide your answers to the questions on the accompanying **answer sheet** by blackening out the corresponding squares. Please carefully read the following instructions regarding the answer sheet:

- During the exam, you may use pencil and eraser to mark and edit your answers on the answer sheet. After the time is up, we will collect the questions pack. Make sure that you marked all your answers by then, at least with a pencil. After collecting the questions pack, we provide you with additional **10** minutes of time to blacken out the squares on the answer sheet with a **black pen**.
- As the answer sheet will be graded by a computer, please **make sure to blacken out the whole square. Do not use ticks or crosses**.
- Nothing written on the pages of the question pack will be collected or marked. **Only the separate answer sheet with the filled squares will be marked**.
- Please make sure that your answer sheet is clean and do not write anything on the answer sheet except the squares you blackened out. We reserve the right to classify answers as wrong without further consideration if the sheet is filled out ambiguously.

Collaboration on the exam is strictly forbidden. You are allowed a summary of *two* A4 pages and a simple, non-programmable calculator. The use of any other helping material or collaboration will lead to being excluded from the exam and subjected to disciplinary measures by the ETH Zurich disciplinary committee.

Question Types In this exam, questions award **1, 2, 3, or 4** points if answered correctly, depending on the difficulty of the question. You will encounter the following question types:

- **True or False.**
- **Multiple choice questions with a *single* correct answer.**

  These multiple choice questions have **exactly one** correct choice.
- **Multiple choice questions with *multiple* correct answers.**

  These multiple choice questions are marked with a "♣" and have **one or more** correct choices (at least one choice is correct). You will receive full points if you have chosen **all** the correct choices and **none** of the wrong choices. You will get zero points otherwise.

There are no negative points in this exam, so **0 points** are awarded if a question is answered wrongly or not attempted. The sections and questions are not ordered by difficulty. Thus, if you find a question too difficult, it may make sense to get back to it in the end.

Questions in the same section might be related. In addition, some questions require information from previous parts. Hence, if you intend to skip some questions, make sure that you read all notes and details between the questions.

The notation $\mathcal{N}(\mu, \sigma^2)$ denotes the Gaussian random distribution with mean μ and variance σ^2. For example, $\mathcal{N}(3, 4)$ designates the Gaussian distribution with mean 3 and standard deviation 2. $\log(\cdot)$ denotes the natural logarithm.

Good luck!

1 Bayesian Linear Regression (9 points)

Question 1 *(1 point)*
Bayesian Linear Regression does not allow to separate epistemic from aleatoric uncertainty.

A True ■ False

Solution:
False. Slide 32. BLR allows to separate epistemic from aleatoric uncertainty.

Question 2 *(1 point)*
Assuming noise is (cond.) i.i.d. Gaussian, and the prior on the model parameters is Gaussian, ridge regression can be understood as finding the Maximum Likelihood (MLE) parameter estimate for a linear regression problem.

A True ■ False

Solution:
False. Slide 14. Under the same assumptions ridge regression can be undestood as finding the Maximum A Posteriori (MAP) parameter estimate.

Question 3 *(2 points)*
We consider a generalized Bayesian Linear Regression. We are given a dataset of inputs $\mathbf{X} \in \mathbb{R}^{n \times d}$, and $\mathbf{y} \in \mathbb{R}^n$. Consider the model

$$\mathbf{y} = \mathbf{X}\mathbf{w} + \epsilon$$

Unlike the standard setting, we assume that the noise is correlated with a covariance matrix $\mathbf{\Sigma}_n \in \mathbb{R}^{n \times n}$, $\epsilon \sim \mathcal{N}(0, \mathbf{\Sigma}_n)$. We place a prior on the weights with a non-zero mean $\mathbf{m}_p \in \mathbb{R}^d$ and a covariance matrix $\mathbf{\Sigma}_p \in \mathbb{R}^{d \times d}$, that is, $\mathbf{w} \sim \mathcal{N}(\mathbf{m}_p, \mathbf{\Sigma}_p)$. What is the marginal distribution of $\mathbf{y}$ given inputs $\mathbf{X}$?

■ $p(\mathbf{y} \mid \mathbf{X}) = \mathcal{N}(\mathbf{X}\mathbf{m}_p, \mathbf{X}\mathbf{\Sigma}_p\mathbf{X}^T + \mathbf{\Sigma}_n)$

B $p(\mathbf{y} \mid \mathbf{X}) = \mathcal{N}(\mathbf{0}, \mathbf{X}\mathbf{\Sigma}_p\mathbf{X}^T + \mathbf{\Sigma}_n)$

C $p(\mathbf{y} \mid \mathbf{X}) = \mathcal{N}(\mathbf{X}\mathbf{m}_p, (\mathbf{\Sigma}_p^{-1} + \mathbf{X}^T\mathbf{\Sigma}_\mathbf{n}^{-1}\mathbf{X})^{-1})$

D $p(\mathbf{y} \mid \mathbf{X}) = \mathcal{N}(\mathbf{X}\mathbf{m}_p, \mathbf{\Sigma}_p + \mathbf{\Sigma}_n)$

Solution:
A. Slide 19. We know that $\mathbf{w} \sim \mathcal{N}(\mathbf{m}_p, \mathbf{\Sigma}_p)$, and $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}_n)$. Since $\mathbf{y}$ is an affine transformation of $\mathbf{w}$, and ϵ it is also Gaussian. The mean is:

$$\mathbb{E}[\mathbf{y}] = \mathbf{X}\mathbb{E}[\mathbf{w}] + \mathbb{E}[\epsilon] = \mathbf{X}\mathbf{m}_p + \mathbf{0} = \mathbf{X}\mathbf{m}_p$$

The covariance is:

$$\text{Var}[\mathbf{y}] = \mathbf{X}\text{Var}[\mathbf{w}]\mathbf{X}^T + \text{Var}[\epsilon] = \mathbf{X}\mathbf{\Sigma}_p\mathbf{X}^T + \mathbf{\Sigma}_n$$

Question 4 *(2 points)*
Assume that the prior over the bias for flipping a coin is distributed as $\text{Beta}(\theta; \alpha, \beta)$. We model the likelihood of the outcome from the coin flips with a binomial distribution as $k \mid \theta \sim \text{Bin}(n, \theta)$ where θ is the probability of heads. Let $n_H = k$ be the number of heads, and $n_T = n - k$ the number of tails in the binomial trial. Then the posterior distribution of the coin given the binomial trials is distributed as $\theta \mid k \sim \text{Beta}(\alpha + n_T, \beta + n_H)$.

Hint: $\text{Bin}(k; n, \theta) = \binom{n}{k}\theta^k(1-\theta)^{n-k}$, and $\text{Beta}(\theta; \alpha, \beta) \propto \theta^{\alpha-1}(1-\theta)^{\beta-1}$.

A True ■ False

Solution:
False. Script page 19. Using Bayes' rule:

$$\begin{aligned} p(\theta \mid k) &\propto p(k \mid \theta)p(\theta) \\ &\propto \theta^{n_H}(1-\theta)^{n_T}\theta^{\alpha-1}(1-\theta)^{\beta-1} \\ &= \theta^{\alpha+n_H-1}(1-\theta)^{\beta+n_T-1} \end{aligned}$$

Hence, the posterior distribution is $\theta \mid k \sim \text{Beta}(\alpha + n_H, \beta + n_T)$.

Question 5 *(1 point)*
Bayesian linear regression can be implemented online where the data $(\mathbf{x} \in \mathbb{R}^d, y \in \mathbb{R})$ is coming one-by-one in an i.i.d. stream, and the posterior distribution is updated recursively. The algorithm can be implemented with $\mathcal{O}(d)$ memory.

A True ■ False

Solution:
False. Homework 1 ex.4. The algorithm requires $\mathcal{O}(d^2)$ space due to the covariance matrix.

Question 6 *(2 points)*
Let $(X, Y) \sim \mathcal{N}(0, \Sigma)$, where $\Sigma = \begin{pmatrix} 1 & \alpha \\ \alpha & 1 \end{pmatrix}$, and $\alpha \in \mathbb{R}, |a| \leq 1$. Then random vectors $U = X + Y$ and $V = X - Y$ are independent if and only if $\alpha = 0$.

A True ■ False

Solution:
False. Slide 50. Gaussians are independent if and only if they are uncorrelated. We compute:

$$\begin{aligned} \text{Cov}(U, V) &= \text{Cov}(X + Y, X - Y) \\ &= \text{Cov}(X, X) - \text{Cov}(Y, Y) + \text{Cov}(Y, X) - \text{Cov}(X, Y) \\ &= 1 - 1 + \alpha - \alpha = 0 \end{aligned}$$

Hence, it does not depend on α.

2 Gaussian Processes (13 points)

2.1 Kernels (7 points)

Question 7 *(2 points)* Let $k : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be a valid kernel, then

$$g(x, x') \triangleq \int_{-\infty}^{\infty} k(x + u, x' + u) \exp\left(-u^2\right) \mathrm{d}u$$

is a valid kernel. You may assume that the integral exists.

■ True B False

Solution:
True. A valid kernel (covariance) function must satisfy *symmetry* and *positive semi-definiteness* (see lecture 3, slide 16):

- Symmetry of the kernel k requires $k(x, x') = k(x', x)$ for all x and $x' \in \mathbb{R}$.
- For positive semi-definiteness of the kernel k, the corresponding kernel matrix $\mathbf{K}_{AA}$ must be positive semi-definite for every $A \subset \mathbb{R}$.

Step 1. Symmetry is easy to check. Note that

$$g(x, x') = \int_{-\infty}^{\infty} k(x+u, x'+u) \exp\left(-u^2\right) \mathrm{d}u = \int_{-\infty}^{\infty} k(x'+u, x+u) \exp\left(-u^2\right) \mathrm{d}u = g(x', x),$$

where we used the symmetry of k.

Step 2. Positive semi-definiteness of g follows from the fact that k is a valid kernel and the linearity of the integral. For arbitrary but fixed $n \in \mathbb{N}$ and $A = \{x_1, ..., x_n\} \subset \mathbb{R}$, we denote the shifted kernel matrix by

$$\mathbf{K}_{AA}(u) \triangleq \left[k(x_i + u, x_j + u)\right]_{i,j=1}^{n} \qquad u \in \mathbb{R}.$$

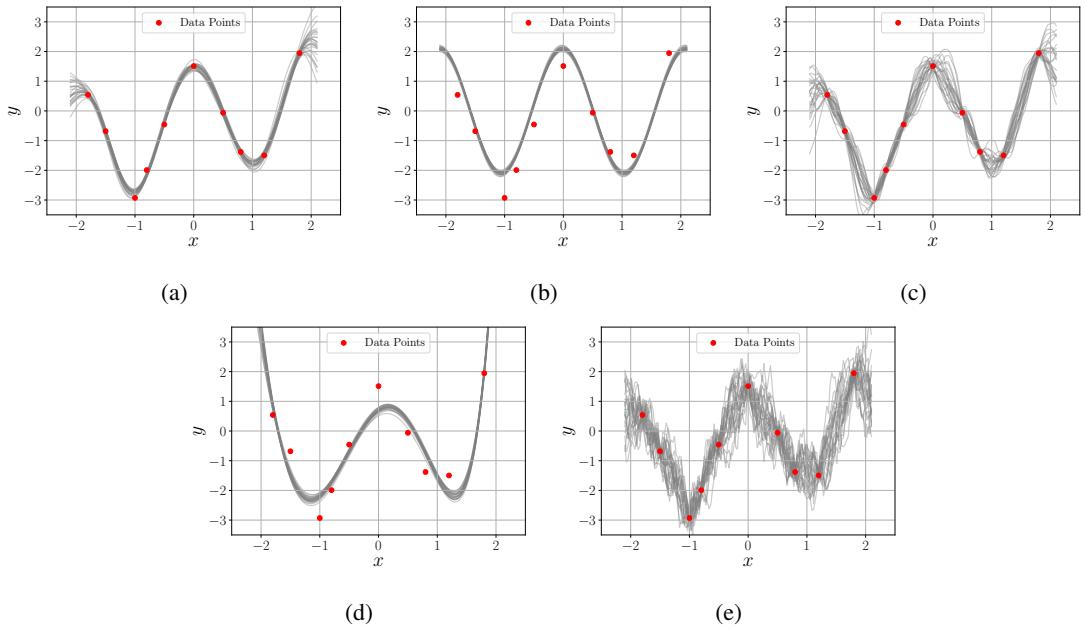
Since k is a kernel, $\mathbf{K}_{AA}(u)$ is positive semi-definite for every $u \in \mathbb{R}$. Now consider the quadratic form of the matrix $\mathbf{G}_{AA}$ for any $\boldsymbol{z} \in \mathbb{R}^n$:

$$\begin{aligned}
\boldsymbol{z}^T \mathbf{G}_{AA} \boldsymbol{z} &= \boldsymbol{z}^T \left(\int_{-\infty}^{\infty} \mathbf{K}_{AA}(u) \exp\left(-u^2\right) \mathrm{d}u \right) \boldsymbol{z} \\
&= \int_{-\infty}^{\infty} \underbrace{\boldsymbol{z}^T \mathbf{K}_{AA}(u) \boldsymbol{z}}_{\geq 0} \exp\left(-u^2\right) \mathrm{d}u \\
&\geq 0,
\end{aligned}$$

which establishes positive semi-definiteness of g. Interchanging the finite quadratic form and the integral is justified by Tonelli's / Fubini's theorem.

Suppose we are given a dataset $\mathcal{D} = \{(x_1, y_1), ..., (x_n, y_n)\}$ of n observations. We fit Gaussian processes with zero mean and the following five kernels to the data $\mathcal{D}$.

- $k_1(x, x') = (1 + xx')^5$
- $k_2(x, x') = \cos\left(3\,|x - x'|\right)$
- $k_3(x, x') = \exp\left(-2\,(x - x')^2\right)$
- $k_4(x, x') = \exp\left(-2\,|x - x'|\right)$



(a) (b) (c)

(d) (e)

Figure 1: Samples of posterior Gaussian processes for different kernels k. The red dots represent the data $\mathcal{D}$.

- $k_5(x, x') = \left(1 + 2\sqrt{3}\,|x - x'|\right) \cdot \exp\left(-2\sqrt{3}\,|x - x'|\right)$

Representative samples from posterior Gaussian processes are depicted in Fig. 1. Match each of the plots in Fig. 1 to the corresponding kernel. There is a one-to-one correspondence between the kernel functions $\{k_1, ..., k_5\}$ and the plots in Fig. 1.

Question 8 *(1 point)* Which plot corresponds to the kernel k_1?

A Plot (a) B Plot (b) C Plot (c) ■ Plot (d) E Plot (e)

Question 9 *(1 point)* Which plot corresponds to the kernel k_2?

A Plot (a) ■ Plot (b) C Plot (c) D Plot (d) E Plot (e)

Question 10 *(1 point)* Which plot corresponds to the kernel k_3?

■ Plot (a) B Plot (b) C Plot (c) D Plot (d) E Plot (e)

Question 11 *(1 point)* Which plot corresponds to the kernel k_4?

A Plot (a) B Plot (b) C Plot (c) D Plot (d) ■ Plot (e)

Question 12 *(1 point)* Which plot corresponds to the kernel k_5?

A Plot (a) B Plot (b) ■ Plot (c) D Plot (d) E Plot (e)

Solution:
Since these questions are interconnected, we provide a single answer to justify the assignment.

(1) Polynomial kernel
k_1 is a polynomial kernel of degree 5. Polynomial functions grow rapidly as x moves away from the origin. It is the only non-stationary kernel in the set since its value depends on the product xx' rather than the distance between x and x' (lecture 3, slide 23). Consequently, the variance grows rapidly as inputs move away from the origin. Plot (d) is the only plot exhibiting this divergence at the boundaries.

(2) Cosine kernel
The cosine kernel k_2 is a periodic function, leading to strictly periodic posterior sample paths with no decay (lecture 3, slide 18). Plot (b) is the only plot showing a perfectly repeating structure.

The following three kernels k_3, k_4, k_5 are members (or limiting cases) of the generalized Matérn class (lecture 3, slide 21). They differ primarily in their smoothness parameter ν, which controls the degree of differentiability of the resulting Gaussian process posterior.

(3) RBF kernel
The squared exponential kernel (also known as the RBF kernel) corresponds to a Matérn kernel with $\nu \to \infty$ (lecture 3, slide 21). Its sample paths are infinitely differentiable, resulting in extremely smooth curves. This corresponds to plot (a).

(4) Exponential kernel
The exponential kernel is a Matérn kernel with $\nu = 1/2$. Its sample paths are continuous but nowhere differentiable (lecture 3, slide 21), leading to rough, jagged paths. This corresponds to plot (e).

(5) Matérn kernel
The Matérn-3/2 kernel produces functions that are once differentiable (lecture 3, slide 21). Visually, this places it in between the extreme roughness of the exponential kernel k_5 and the extreme smoothness of the RBF kernel k_4. The samples in plot (c) are smooth but exhibit subtle local roughness compared to (a), making it the correct match.

2.2 Bayesian Model Selection (6 Points)

Consider a regression model where the target $y \in \mathbb{R}$ is generated from inputs $x \in \mathbb{R}$ according to:

$$y = f(x) + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2),$$

for an unknown function f. Assume the noise variance is known to be $\sigma^2 = 1$. We place a Gaussian Process prior on the function f, denoted as $f \sim \text{GP}(0, k)$, using a cosine kernel with an angular frequency parameter ω:

$$k(x, x') = \cos\big(\omega\,|x - x'|\big).$$

We observe a dataset $\mathcal{D}$ consisting of two points:

$$(x_1, y_1) = (0, 1) \qquad (x_2, y_2) = (\pi/2, 0).$$

To tune the angular frequency ω of the kernel, we maximize the marginal likelihood $p(y_1, y_2 \mid x_1, x_2, \omega)$.

Question 13 *(4 points)* Which of the following sets contains *all* values of ω that represent stationary points (critical points) of the log-likelihood function $\log p(y_1, y_2 \mid x_1, x_2, \omega)$?

Hint I: The marginal log-likelihood is given by

$$\log p(y_1, y_2 \mid x_1, x_2, \omega) = -\frac{1}{2}\,\boldsymbol{y}^T \big(\mathbf{K} + \sigma^2\mathbf{I}\big)^{-1}\boldsymbol{y} - \frac{1}{2}\log\det(\mathbf{K} + \sigma^2\mathbf{I}) + \text{const.}$$

Hint II: The determinant of a 2×2 *matrix is given by*

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc.$$

Hint III: The inverse of an invertible 2×2 *matrix is given by*

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

A $\{\omega = 2k \text{ for } k \in \mathbb{Z}\}$

B $\{\omega = 2k \text{ for } k \in \mathbb{N}_{\geq 0}\}$

C $\{\omega = 2k + 1 \text{ for } k \in \mathbb{Z}\}$

■ $\{\omega = k \text{ for } k \in \mathbb{Z}\}$

E $\{\omega = \pi k \text{ for } k \in \mathbb{Z}\}$

Solution:
D. We define the vector $\boldsymbol{y} = (y_1, y_2)^T = (1, 0)^T$ as well as the kernel matrix $\mathbf{K}_{ij} = k(x_i, x_j)$. Following the first hint (or lecture 4, slide 12), we have

$$\log p(y_1, y_2 \,|\, x_1, x_2, \omega) = -\frac{1}{2} \boldsymbol{y}^T \left(\mathbf{K} + \sigma^2 \mathbf{I}\right)^{-1} \boldsymbol{y} - \frac{1}{2} \log \det(\mathbf{K} + \sigma^2 \mathbf{I}) + \text{const.}$$

Using the definition of the cosine kernel and the fact that $\sigma^2 = 1$, we obtain

$$\mathbf{K} + \sigma^2 \mathbf{I} = \begin{pmatrix} 2 & \cos(\pi\omega/2) \\ \cos(\pi\omega/2) & 2 \end{pmatrix}.$$

The formula for the determinant is given in the second hint. The inverse of this matrix can be computed with the help of the third hint. With these insights, we can further simplify the log-likelihood to obtain

$$\log p(y_1, y_2 \,|\, x_1, x_2, \omega) = -\frac{1}{2} \frac{2}{4 - \cos^2(\pi\omega/2)} - \frac{1}{2} \log \left[4 - \cos^2(\pi\omega/2)\right] + \text{const.}$$

To find extreme values for ω, we set the derivative with respect to ω to zero.

$$\begin{aligned} \frac{\partial}{\partial \omega} \log p(y_1, y_2 \,|\, x_1, x_2, \omega) &= \frac{1}{2} \frac{2\pi \cos(\pi\omega/2) \sin(\pi\omega/2)}{\left[4 - \cos^2(\pi\omega/2)\right]^2} - \frac{1}{2} \frac{\pi \cos(\pi\omega/2) \sin(\pi\omega/2)}{4 - \cos^2(\pi\omega/2)} \\ &= \frac{1}{2} \frac{\left[\cos^2(\pi\omega/2) - 2\right] \pi \cos(\pi\omega/2) \sin(\pi\omega/2)}{\left[4 - \cos^2(\pi\omega/2)\right]^2} \\ &\stackrel{!}{=} 0. \end{aligned}$$

Note that the equality holds if and only if either $\cos(\pi\omega/2) = 0$ or $\sin(\pi\omega/2) = 0$. Recall that

$$\begin{aligned} \cos(\pi\omega/2) = 0 \quad &\iff \quad \omega \in \{2k + 1 : k \in \mathbb{Z}\} \\ \sin(\pi\omega/2) = 0 \quad &\iff \quad \omega \in \{2k : k \in \mathbb{Z}\}. \end{aligned}$$

Therefore, we conclude that the set $\{\omega = k \text{ for } k \in \mathbb{Z}\}$ contains all critical points of the log-likelihood.

In the previous question, we established that the log-likelihood $\log p(y_1, y_2 \,|\, x_1, x_2, \omega)$ has a countably infinite set of critical points. Upon closer inspection, we see that these points alternate between local maxima and minima, leaving us with an infinite number of Maximum Likelihood Estimators (MLEs). To resolve this non-uniqueness, we place a hyperprior on ω. This acts as a form of regularization, allowing us to single out a unique MAP estimate.

Question 14 *(2 points)* Let us impose a hyperprior on ω with the following probability distribution:

$$\mathbb{P}(\omega = k) = \begin{cases} \frac{1}{2^{k+1}} & k \in \mathbb{N}_{\geq 0} \\ 0 & \text{otherwise.} \end{cases}$$

Calculate the MAP estimate $\hat{\omega}$, defined as the solution to the optimization problem:

$$\hat{\omega} = \arg\max_{\omega} \log p(\omega \mid x_1, x_2, y_1, y_2).$$

Hint: You may use the approximations:

- $\log 2 \approx 0.69$
- $\log 3 \approx 1.10$
- $\log 4 \approx 1.39$

A $\hat{\omega} = 1$ ■ $\hat{\omega} = 0$ C $\hat{\omega} = 2$ D $\hat{\omega} = 3$ E $\hat{\omega} = -1$

Solution:
B. The MAP estimate is given by $\hat{\omega} = 0$.

Intuition. The hyperprior assigns a decaying weight to the parameters $\omega \in \mathbb{N}_{\geq 0}$ (and zero weight elsewhere), favoring smaller values over larger ones. From the previous problem, we know that the log-likelihood depends on ω only through $\cos^2(\pi\omega/2)$. Because of the cosine's periodicity, all even values of ω yield the same likelihood, and all odd values yield another common likelihood. Consequently, it suffices to consider only $\omega \in \{0, 1\}$, as larger values receive smaller prior weights. The final step is then simply to compare these two options to determine which gives the higher posterior.

Detailed Solution. From lecture 4, slide 17, we know that

$$\arg\max_{\omega} p(\omega \mid x_1, x_2, y_1, y_2) = \arg\max_{\omega} p(y_1, y_2 \mid x_1, x_2, \omega)\, p(\omega).$$

This can easily be verified by applying Bayes' rule to the posterior. Given that the support of the hyperprior is restricted to $\omega \in \mathbb{N}_{\geq 0}$, the optimization problem reduces to:

$$\hat{\omega} = \arg\max_{\omega \in \mathbb{N}_{\geq 0}} \Big[\underbrace{\log p(y_1, y_2 \mid x_1, x_2, \omega) + \log p(\omega)}_{\mathcal{L}_{\text{MAP}}(\omega)} \Big].$$

The natural logarithm of the hyperprior is given by

$$\log p(\omega) = -(\omega + 1) \log 2, \qquad \omega \in \mathbb{N}_{\geq 0}.$$

Next, we recall the log-likelihood from the previous exercise, which reads as

$$\log p(y_1, y_2 \mid x_1, x_2, \omega) = -\frac{1}{2} \frac{2}{4 - \cos^2(\pi\omega/2)} - \frac{1}{2} \log \left[4 - \cos^2(\pi\omega/2)\right] + c,$$

for some constant $c \in \mathbb{R}$ independent of ω. Note that $\cos^2(\pi\omega/2) = 1$ for even ω and $\cos^2(\pi\omega/2) = 0$ for odd ω. Therefore, for $\omega \in \mathbb{N}_{\geq 0}$, we have

$$\mathcal{L}_{\text{MAP}}(\omega) = \begin{cases} -\frac{1}{3} - \frac{1}{2}\log(3) - (\omega + 1)\log 2 + c & \text{if } \omega \text{ is even} \\ -\frac{1}{4} - \frac{1}{2}\log(4) - (\omega + 1)\log 2 + c & \text{if } \omega \text{ is odd.} \end{cases}$$

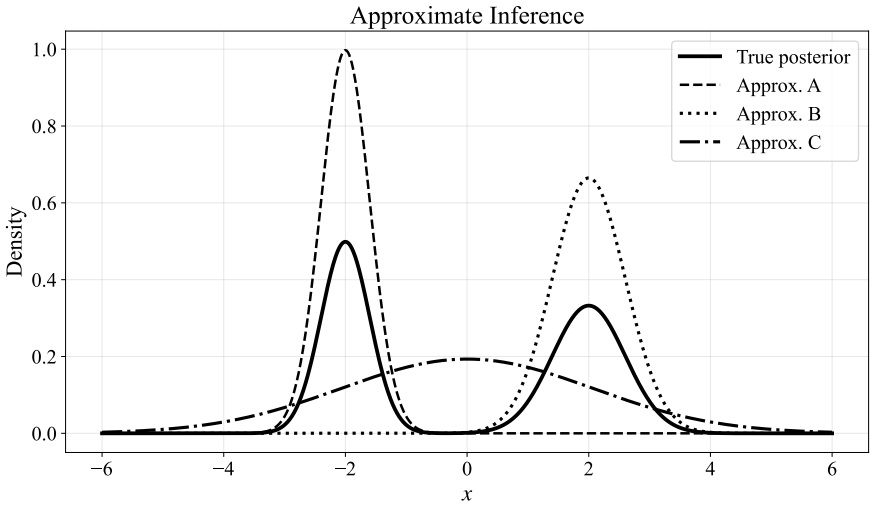
Because the prior penalty grows monotonically, we can see that $\omega = 0$ is the maximizer among the even candidates and $\omega = 1$ is the maximizer among the odd candidates. With the approximations in the hint, we obtain

$$\mathcal{L}_{\text{MAP}}(\omega = 0) \approx -0.33 - 0.5 \cdot 1.10 - 1 \cdot 0.69 + c = -1.57 + c$$
$$\mathcal{L}_{\text{MAP}}(\omega = 1) \approx -0.25 - 0.5 \cdot 1.39 - 2 \cdot 0.69 + c = -2.325 + c.$$

Hence, we can deduce that $\hat{\omega} = 0$ is the correct solution.

3 Variational Inference & Diffusion (12 points)

3.1 Variational Inference (7 points)



In the plot above, the solid black curve is a bimodal posterior distribution $p(x \mid y)$. The three other curves (A, B, C) are Gaussian approximations produced by three different approximate inference methods. Match each curve to the corresponding method:

- Laplace approximation;
- Variational inference with a Gaussian $q(x) = \mathcal{N}(\mu, \sigma^2)$ minimizing reverse-KL, $\mathrm{KL}(q\|p)$;
- Variational inference with a Gaussian $q(x) = \mathcal{N}(\mu, \sigma^2)$ minimizing forward-KL, $\mathrm{KL}(p\|q)$.

Question 15 *(1 point)* Which method corresponds to approximation A?

■ Laplace B Reverse-KL C Forward-KL

Question 16 *(1 point)* Which method corresponds to approximation B?

A Laplace ■ Reverse-KL C Forward-KL

Question 17 *(1 point)* Which method corresponds to approximation C?

A Laplace B Reverse-KL ■ Forward-KL

Solution:
A, B, C.
From Lecture 5, slide 8-9 and slide 22-24.
Curve A is a narrow Gaussian centered on one posterior mode, which is characteristic of the Laplace approximation (local second-order approximation around a MAP mode).
Curve B corresponds to Gaussian variational inference minimizing $\mathrm{KL}(q\|p)$. This objective is mode-seeking. Which mode is selected depends on the initialization of the variational parameters. Since Laplace is centered on the posterior mode, only curve B is left for reverse-KL.
Curve C is a broad Gaussian that covers both modes, consistent with a reverse-KL / mass-covering fit (e.g., moment matching), which tends to avoid missing probability mass even if it places density in low-probability regions between modes.

Question 18 ♣ *(4 points)* Consider a 1D Bayesian model

$$y = w + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \sigma_n^2),$$

with prior $w \sim \mathcal{N}(0, \sigma_p^2)$. We approximate the posterior $p(w \mid y)$ by a Gaussian variational distribution $q(w) = \mathcal{N}(m, s^2)$. The ELBO is

$$\mathcal{L}(m, s^2) = \mathbb{E}_q[\log p(y \mid w)] - \mathrm{KL}(q\|p(w)).$$

Let m_{opt} and s^2_{opt} be the maximizers of $\mathcal{L}$.
Hint: A 1D Gaussian $x \sim \mathcal{N}(\mu, \sigma^2)$ has the following probability density function

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}\right).$$

The variance of a random variable X can be calculated by $Var(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2$.
The KL divergence between two 1D Gaussians $p_0 = \mathcal{N}(\mu_0, \sigma_0^2)$ and $p_1 = \mathcal{N}(\mu_1, \sigma_1^2)$ is

$$\mathrm{KL}(p_0\|p_1) = \log\frac{\sigma_1}{\sigma_0} + \frac{\sigma_0^2 + (\mu_0 - \mu_1)^2}{2\sigma_1^2} - \frac{1}{2}.$$

Which of the following statements are **true**? **Mark all that apply.**

- ■ As $\sigma_n^2 \to 0$, $s^2_{\text{opt}} \to 0$.
- ■ As $\sigma_n^2 \to \infty$, $m_{\text{opt}} \to 0$.
- C As $\sigma_n^2 \to \infty$, $s^2_{\text{opt}} \to 0$.
- D Increasing σ_p^2 always decreases s^2_{opt}.

Solution:
Correct: A, B.
From Lecture 5, slides 28–33, we know the ELBO form for Gaussian likelihood + Gaussian prior, and how to compute the expectations and KL analytically.
We have

$$\log p(y \mid w) = -\frac{1}{2}\log(2\pi\sigma_n^2) - \frac{1}{2\sigma_n^2}(y-w)^2.$$

Taking expectation under q,

$$\mathbb{E}_q[(y-w)^2] = \mathrm{Var}_q(w) + (y - \mathbb{E}_q[w])^2 = s^2 + (y-m)^2.$$

Thus,

$$\mathbb{E}_q[\log p(y \mid w)] = \text{const} - \frac{1}{2\sigma_n^2}\big(s^2 + (y-m)^2\big).$$

For $q(w) = \mathcal{N}(m, s^2)$ and $p(w) = \mathcal{N}(0, \sigma_p^2)$,

$$\mathrm{KL}(q\|p) = \frac{1}{2}\left(\frac{s^2}{\sigma_p^2} + \frac{m^2}{\sigma_p^2} - 1 - \log\frac{s^2}{\sigma_p^2}\right).$$

Ignoring additive constants, the objective to maximize is

$$\tilde{\mathcal{L}}(m, s^2) = -\frac{1}{2\sigma_n^2}\big(s^2 + (y-m)^2\big) - \frac{1}{2}\left(\frac{s^2+m^2}{\sigma_p^2} - 1 - \log\frac{s^2}{\sigma_p^2}\right).$$

Differentiate w.r.t. m:

$$\frac{\partial\tilde{\mathcal{L}}}{\partial m} = -\frac{m-y}{\sigma_n^2} - \frac{m}{\sigma_p^2}.$$

Setting this equal to zero,

$$\boxed{m_{\text{opt}} = \frac{\sigma_p^2}{\sigma_p^2+\sigma_n^2}\,y}.$$

Differentiate w.r.t. s^2:

$$\frac{\partial\tilde{\mathcal{L}}}{\partial s^2} = -\frac{1}{2\sigma_n^2} - \frac{1}{2\sigma_p^2} + \frac{1}{2s^2}.$$

Setting this to zero:

$$\boxed{s^2_{\text{opt}} = \left(\frac{1}{\sigma_p^2} + \frac{1}{\sigma_n^2}\right)^{-1}}.$$

A: If $\sigma_n^2 \to 0$, the posterior concentrates, so

$$s^2_{\text{opt}} = \left(\frac{1}{\sigma_p^2} + \frac{1}{\sigma_n^2}\right)^{-1} \to 0.$$

B: If $\sigma_n^2 \to \infty$, then $m_{\text{opt}} = s^2_{\text{opt}}\frac{y}{\sigma_n^2} \to 0$. Interpretation: data is uninformative $\to$ revert to prior mean.
C: False: as $\sigma_n^2 \to \infty$,

$$s^2_{\text{opt}} \to \sigma_p^2,$$

not to 0.
D: False: increasing σ_p^2 increases s^2_{opt}.

3.2 Diffusion (5 Points)

Question 19 *(1 point)* Diffusion is a form of supervised learning that requires labeled data $\{x^{(i)}, y^{(i)}\}_{i=1}^n$.

A True ■ False

Solution:
False. Lecture 8 Slide 3. Diffusion is an unsupervised learning problem.

Question 20 *(1 points)*
A longer diffusion process in the DDPM model, i.e., larger T, makes sampling new datapoints computationally more expensive.

■ True B False

Solution:
True. Detailed in Homework 4 Exercise 6b.

Question 21 *(1 points)*
The training time per SGD iteration increases with larger time-horizon T in the DDPM training procedure as covered in the lecture.

A True ■ False

Solution:
False. Lecture 8 Slide 42. Algorithm 1's runtime is independent of T as it samples a random timestep t for each SGD step using loss L_t^ϵ.

Question 22 *(2 points)*
Consider a diffusion on the domain $\mathbb{R}$ with the following one-step forward-process distribution

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}x_{t-1}, 2\beta_t)$$

where $\beta_t \in (0, 1]$ is a monotonically increasing sequence. What is the marginal distribution $q(x_t|x_0)$ in the limit when $t \to \infty$?
Hint: The distribution of the convolution of gaussian random variables can be calculated as follows.

$$\mathcal{N}(y; a\mu, b + a^2\sigma^2) = \int \mathcal{N}(x; \mu, \sigma^2)\mathcal{N}(y; ax, b)dx$$

where $\mathcal{N}(x; \mu, \sigma^2)$ and $\mathcal{N}(y; ax, b)$.

A $q(x_t|x_0) = \mathcal{N}(2, 1)$
B $q(x_t|x_0) = \mathcal{N}(0, 1)$
■ $q(x_t|x_0) = \mathcal{N}(0, 2)$
D $q(x_t|x_0) = \mathcal{N}(0, 4)$

Solution:
C. Following Homework 4 Exercise 5, the marginal $q(x_t|x_0)$ can be derived via induction. Let $\overline{\alpha}_t = \Pi_{i=1}^{t}(1-\beta_i)$. Then base case is given as

$$q(x_1|x_0) = \mathcal{N}(x_1; \sqrt{1-\beta_t}x_0, 2\beta_1) = \mathcal{N}(x_1; \sqrt{\overline{\alpha}_1}x_0, 2(1-\overline{\alpha}_1))$$

We use the convolution of Gaussians to show the induction step

$$\begin{aligned}
q(x_{t+1}|x_0) = &\int q(x_{t+1}, x_t|x_0)dx_t \\
&\int q(x_t|x_0)q(x_{t+1}|x_t)dx_t \\
&\int \mathcal{N}(x_t; \sqrt{\overline{\alpha}_t}x_0, 2(1-\overline{\alpha}_t))\mathcal{N}(x_{t+1}; \sqrt{1-\beta_{t+1}}x_t, 2\beta_{t+1})dx_t \\
&= \mathcal{N}(x_{t+1}; \sqrt{(1-\beta_{t+1})\overline{\alpha}_t}x_0, 2\beta_{t+1} + 2(1-\beta_{t+1})(1-\overline{\alpha}_t)) \\
&= \mathcal{N}(x_{t+1}; \sqrt{\overline{\alpha}_{t+1}}x_0, 2(1-\overline{\alpha}_{t+1})).
\end{aligned}$$

Since β_t is monotonically increasing, $\overline{\alpha}_t \to 0$ as $t \to \infty$, therefore, the marginal distribution of $q(x_t|x_0) = \mathcal{N}(0, 2)$ in the limit.

4 Active Learning & Bayesian Optimization (8 points)

Question 23 *(1 point)* In a Gaussian Process $p(f) = \text{GP}(f; \mu, k)$ with Gaussian observation noise $y = f(x) + \epsilon, \epsilon \sim \mathcal{N}(0, \sigma_n^2)$, the mutual information $I(f(x); y)$ increases as the noise variance σ_n^2 increases.

[A] True ■ False

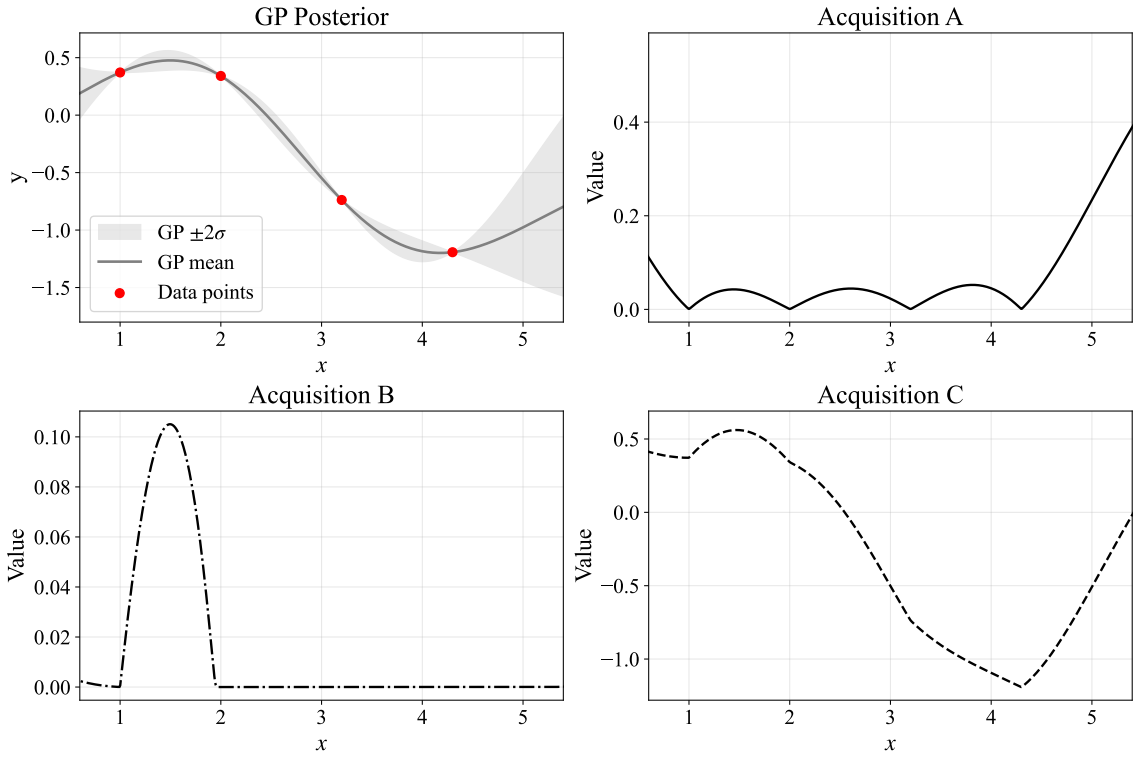
Solution:
False. From lecture 7, slide 11. Higher noise reduces the informativeness of an observation, so mutual information decreases.

$$I(f; y) = \frac{1}{2} \log \left(1 + \frac{k(x, x)}{\sigma_n^2}\right).$$

Question 24 *(1 point)* Thompson Sampling with a Gaussian process model selects a point x by:

[A] Maximizing the posterior mean.
[B] Minimizing posterior variance.
■ Sampling a function from the Gaussian process posterior and choosing the maximizer of that sample.
[D] Selecting the point with highest expected improvement.

Solution:
C. From lecture 7, slide 51.



Consider the figure above showing a Gaussian process posterior together with three acquisition functions labeled A, B, and C. Let $\mu(x)$ and $\sigma(x)$ denote the GP predictive mean and standard deviation, and let f_{best} be the best observed function value so far.
The following acquisition functions are defined as:

- Uncertainty sampling (US):
$$a_{\text{US}}(x) = \sigma(x)$$
- Upper confidence bound (UCB) with $\beta > 0$:
$$a_{\text{UCB}}(x) = \mu(x) + \beta\sigma(x)$$
- Expected improvement (EI):
$$a_{\text{EI}}(x) = \mathbb{E}[\max(0, f(x) - f_{\text{best}})]$$

Match each acquisition function to the corresponding curve in the figure.

Question 25 *(1 point)* Which acquisition function corresponds to curve A?

■ US B UCB ($\beta = 2$) C EI

Question 26 *(1 point)* Which acquisition function corresponds to curve B?

A US B UCB ($\beta = 2$) ■ EI

Question 27 *(1 point)* Which acquisition function corresponds to curve C?

A US ■ UCB ($\beta = 2$) C EI

Solution:
A, C, B.
Lecture 7, slide 8-9, 26, 29, 40-42, Homework 04.
Uncertainty sampling depends only on the predictive standard deviation $\sigma(x)$ and therefore highlights regions far from observed data points, independent of the predictive mean; this corresponds to curve A.
Expected improvement is nonzero only in regions where improvement over the current best observed value is likely and accounts for the magnitude of that improvement, leading to sharp, localized peaks near promising locations; this corresponds to curve B.
Upper confidence bound combines exploitation and exploration through $\mu(x) + \beta\sigma(x)$, producing a smooth curve that is large where either the predictive mean or uncertainty is high; this corresponds to curve C.

Question 28 ♣ *(3 points)* Consider a Bayesian binary classification with latent parameter θ. For an input x, denote
$$p_\theta = p(y = 1 \mid x, \theta).$$
Assume an approximate posterior $q(\theta)$, and define the BALD (Bayesian Active Learning by Disagreement) acquisition score
$$I(\theta; y_x) = H(\mathbb{E}_{\theta\sim q}[p_\theta]) - \mathbb{E}_{\theta\sim q}[H(p_\theta)],$$
where $H(p) = -p\log p - (1-p)\log(1-p)$ is the Bernoulli entropy.
Which of the following statements are **true**? **Mark all that apply.**

■ If all posterior samples predict the same probability $p_\theta = p_0$, then the BALD score at x is zero.

[B] The BALD score depends only on the predictive mean $\mathbb{E}_\theta[p_\theta]$, not on its variance.

■ Consider two cases: (A) uncertainty is aleatoric (i.e., $p_\theta = 0.5$); and (B) uncertainty is epistemic (i.e., $p_\theta = 0$ and $p_\theta = 1$ with probability 0.5 each). The BALD scores for cases (A) and (B) are different.

Solution:
A, C.
From Lecture 7, slide 20.
A: If $p_\theta = p_0$ a.s., then $\bar{p} = p_0$ and $\mathbb{E}_\theta[H(p_\theta)] = H(p_0)$, giving

$$I(\theta; y_x) = H(p_0) - H(p_0) = 0.$$

Intuitively, all posterior samples agree, so there is no epistemic uncertainty left; observing y_x cannot reduce uncertainty about θ.
B: In both cases (i) $p_\theta = 0.5$ a.s. and (ii) $p_\theta \in \{0, 1\}$ with equal probability, we have $\bar{p} = 0.5$. But $I(\theta; y_x) = 0$ in (i) and $I(\theta; y_x) = H(0.5) > 0$ in (ii). Thus BALD is not determined by $\bar{p}$ alone.
C: Case (i) above (all $p_\theta = 0.5$) represents pure aleatoric uncertainty: $I(\theta; y_x) = 0$. Case (ii) (half at 0, half at 1) represents pure epistemic uncertainty: $I(\theta; y_x) = H(0.5)$. Thus, BALD distinguishes epistemic from aleatoric uncertainty.

5 MDPs & Reinforcement Learning (18 points)

5.1 Influence of the action cost (6 points)

For the next two questions, consider an MDP with an infinite set of states $S = \{1, 2, 3, \dots\}$ and start state $s = 1$. Each state allows a continuous action $a \in [0, 1]$. For every state s, the transitions are:

$$P(s+1 \mid s, a) = a, \qquad P(s \mid s, a) = 1 - a.$$

The rewards are defined as follows:

$$r(s, a, s+1) = 1 - \lambda a^2, \qquad r(s, a, s) = -\lambda a^2,$$

where $\lambda \geq 0$ represents action cost regularization. The discount factor is $\gamma = 0.5$.

Question 29 *(2 points)* First, assume that there is no action regularization, i.e., $\lambda = 0$. What is the optimal policy $\pi(s)$ in this case?

A $\pi(s) = \frac{1}{s^2}$ B $\pi(s) = \frac{1}{s}$ ■ $\pi(s) = 1$ D $\pi(s) = \frac{1}{2}$

Solution:
C. We get a reward of 1 if we move from state s to state $s + 1$ and a reward of 0 if we don't move. Hence, the optimal policy is to have the probability of moving as high as possible. Hence $\pi(s) = 1$.

Question 30 *(4 points)* Now, assume that we penalize action with $\lambda = 2$. What is the optimal value $V^*(1)$ in this case?

■ $V^*(1) = \frac{1}{4}$ B $V^*(1) = \frac{1}{2}$ C $V^*(1) = 1$ D $V^*(1) = 2$

Solution:
A. Because all states have identical transition dynamics and reward structure, the optimal value is the same in every state. Let $V^* := V^*(s)$ for all $s \in S$.

Bellman optimality equation. For any state s, the Bellman optimality equation is

$$V^* = \max_{a\in[0,1]} \Big[a\big((1-\lambda a^2) + \gamma V^*\big) + (1-a)\big((-\lambda a^2) + \gamma V^*\big)\Big].$$

We first simplify the immediate reward term:

$$a(1-\lambda a^2) + (1-a)(-\lambda a^2) = a - \lambda a^3 - \lambda a^2 + \lambda a^3 = a - \lambda a^2.$$

The terms involving V^* combine as

$$a\gamma V^* + (1-a)\gamma V^* = \gamma V^*.$$

Thus the Bellman equation becomes

$$V^* = \gamma V^* + \max_{a\in[0,1]} \big(a - \lambda a^2\big).$$

Rearranging, we get

$$(1-\gamma)V^* = \max_{a\in[0,1]} f(a), \qquad f(a) := a - \lambda a^2.$$

With $\gamma = 0.5$, this is

$$0.5\, V^* = \max_{a\in[0,1]} f(a).$$

Maximizing $f(a)$ for $\lambda = 2$. Now set $\lambda = 2$:

$$f(a) = a - 2a^2.$$

We differentiate:

$$f'(a) = 1 - 4a.$$

Setting $f'(a) = 0$ gives the unconstrained maximizer:

$$1 - 4a \stackrel{!}{=} 0 \quad \Longrightarrow \quad a^* = \frac{1}{4}.$$

Since $\frac{1}{4} \in (0,1)$, this is a valid interior maximizer for $a \in [0,1]$.
We evaluate $f(a^*)$:

$$f\left(\frac{1}{4}\right) = \frac{1}{4} - 2\left(\frac{1}{4}\right)^2 = \frac{1}{4} - 2\cdot\frac{1}{16} = \frac{1}{4} - \frac{1}{8} = \frac{1}{8}.$$

Compute V^*. From

$$0.5\, V^* = \max_{a\in[0,1]} f(a) = \frac{1}{8},$$

we obtain

$$V^* = 2\cdot\frac{1}{8} = \frac{1}{4}.$$

As $V^*(s)$ is the same for all states, we conclude that $V^*(1) = \frac{1}{4}$.

5.2 True/False questions (5 points)

For the next four questions, assume a finite MDP with discount factor $\gamma \in (0,1)$.

Question 31 *(1 point)* If two deterministic policies π_1, π_2, satisfy $V^{\pi_1}(s) = V^{\pi_2}(s)$ for every state s, then π_1 and π_2 must be identical.

A True ■ False

Solution:
False. We provide a counterexample. Consider an MDP with: $\mathcal{S} = \{s\}, \mathcal{A} = \{a_1, a_2\}$, discount factor $\gamma \in (0, 1)$, and the following dynamics and rewards: $P(s \mid s, a_1) = P(s \mid s, a_2) = 1, r(s, a_1) = r(s, a_2) = 1$. Define two deterministic policies: $\pi_1(s) = a_1, \pi_2(s) = a_2$. Clearly, $\pi_1 \neq \pi_2$ since they choose different actions in state s. For either policy $\pi \in \{\pi_1, \pi_2\}$, the value function at s satisfies $V^\pi(s) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r(s_t, \pi(s_t)) \,\middle|\, s_0 = s\right] = 1 + \gamma V^\pi(s)$, because at every time step the agent receives reward 1 and remains in s. Solving the Bellman equation, $V^\pi(s) = 1 + \gamma V^\pi(s) \implies V^\pi(s)(1 - \gamma) = 1 \implies V^\pi(s) = \frac{1}{1-\gamma}$. Thus, $V^{\pi_1}(s) = \frac{1}{1-\gamma} = V^{\pi_2}(s)$, for the (only) state s, even though $\pi_1 \neq \pi_2$. Hence, it is possible for two different deterministic policies to have identical value functions in every state, so the claim is false.

Question 32 *(1 point)* For any value function V, the greedy policy

$$\pi(x) = \arg\max_a \left[r(x, a) + \gamma \sum_{x'} P(x' \mid x, a)\, V(x') \right]$$

always satisfies $V^\pi = V$.

A True ■ False

Solution:
False. We provide a simple counterexample.
Consider an MDP with:

$$\mathcal{X} = \{x\}, \qquad \mathcal{A} = \{a_1, a_2\},$$

discount factor $\gamma \in (0, 1)$, and the following rewards and transitions:

$$P(x \mid x, a_1) = P(x \mid x, a_2) = 1, \qquad r(x, a_1) = 0, \qquad r(x, a_2) = 1.$$

Now take the (arbitrary) value function

$$V(x) = 0.$$

The greedy policy with respect to V is

$$\pi(x) = \arg\max_{a \in \{a_1, a_2\}} \Big[r(x, a) + \gamma V(x) \Big] = \arg\max_{a \in \{a_1, a_2\}} r(x, a),$$

since $V(x) = 0$ and $P(x \mid x, a) = 1$ imply

$$r(x, a) + \gamma \sum_{x'} P(x' \mid x, a) V(x') = r(x, a) + \gamma V(x) = r(x, a).$$

Thus,

$$r(x, a_1) = 0, \qquad r(x, a_2) = 1 \quad \Rightarrow \quad \pi(x) = a_2.$$

Let us compute the value function V^π of this greedy policy π. Since the process always stays in x and action a_2 yields reward 1 at every time step,

$$\begin{aligned} V^\pi(x) &= \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r(x_t, \pi(x_t)) \,\Big|\, x_0 = x \right] \\ &= 1 + \gamma + \gamma^2 + \cdots \\ &= \frac{1}{1 - \gamma}. \end{aligned}$$

But our original V was $V(x) = 0$, so

$$V^\pi(x) = \frac{1}{1 - \gamma} \neq 0 = V(x).$$

Hence $V^\pi \neq V$, showing that the statement is false: a policy greedy with respect to an arbitrary value function V does *not* in general satisfy $V^\pi = V$. Equality holds only in special cases (e.g. when V is already the value function of a greedy policy or the optimal value function).

Question 33 *(2 points)* Let W, V be two value functions such that

$$W(x) \leq V(x) \quad \text{for all } x.$$

Let B denote the Bellman operator, defined by

$$(BU)(x) \coloneqq \max_a \Big[r(x, a) + \gamma \sum_{x'} P(x' \mid x, a)\, U(x') \Big]$$

for any value function U, where $0 \leq \gamma < 1$. Then it holds that

$$(BW)(x) \leq (BV)(x) \quad \text{for all } x \in \mathcal{X}.$$

■ True B False

Solution:
True Assume $W(x) \le V(x)$ for all $x \in \mathcal{X}$. By definition of B, we have

$$(BW)(x) = \max_a \left[r(x,a) + \gamma \sum_{x'} P(x' \mid x,a)\, W(x') \right],$$

$$(BV)(x) = \max_a \left[r(x,a) + \gamma \sum_{x'} P(x' \mid x,a)\, V(x') \right].$$

Since $W(x') \le V(x')$ for all x' and $P(x' \mid x,a) \ge 0$ with $\sum_{x'} P(x' \mid x,a) = 1$, it follows that for every x and a,

$$\sum_{x'} P(x' \mid x,a)\, W(x') \;\le\; \sum_{x'} P(x' \mid x,a)\, V(x').$$

Multiplying by $\gamma \ge 0$ and adding $r(x,a)$ preserves the inequality:

$$r(x,a) + \gamma \sum_{x'} P(x' \mid x,a)\, W(x') \;\le\; r(x,a) + \gamma \sum_{x'} P(x' \mid x,a)\, V(x').$$

Define

$$Q_W(x,a) \coloneqq r(x,a) + \gamma \sum_{x'} P(x' \mid x,a)\, W(x'), \quad Q_V(x,a) \coloneqq r(x,a) + \gamma \sum_{x'} P(x' \mid x,a)\, V(x').$$

Then $Q_W(x,a) \le Q_V(x,a)$ for all a, and hence taking $\max_a$ on both sides yields

$$(BW)(x) = \max_a Q_W(x,a) \;\le\; \max_a Q_V(x,a) = (BV)(x).$$

Thus $(BW)(x) \le (BV)(x)$ for all $x \in \mathcal{X}$, so the statement is true.

Question 34 *(1 point)* Policy iteration converges to an optimal policy π^* in finite iterations.

■ True B False

Solution:
True. See slides from lecture 9, page 16.

5.3 Applying TD & Q-learning (3 points)

Consider an observed transition (x, a, r, x', a'), with $r = 4.5$. The current estimates are $V(x) = 5$, $Q(x,a) = 2$, and $V(x') = 7$. Use discount $\gamma = 0.5$ and learning rate $\alpha = 0.25$.

Question 35 *(2 points)* The TD(0) update produces the new value:
Hint: TD(λ) learns from a weighted average of n-step returns. For TD(0), the update uses only the one-step return:

$$V(x) \;\leftarrow\; V(x) + \alpha\Big(r + \gamma V(x') - V(x)\Big).$$

A $V(x) = 5$ ■ $V(x) = 5.75$ C $V(x) = 6.5$ D $V(x) = 7$

Solution:
B. We apply the TD(0) update:

$$V(x) \leftarrow V(x) + \alpha\left(r + \gamma V(x') - V(x)\right).$$

Substituting the given values

$$V(x) = 5, \qquad r = 4.5, \qquad \gamma = 0.5, \qquad V(x') = 7, \qquad \alpha = 0.25,$$

we compute the TD target:

$$r + \gamma V(x') = 4.5 + 0.5 \cdot 7 = 4.5 + 3.5 = 8.$$

The TD error is therefore

$$8 - 5 = 3.$$

Applying the update:

$$V(x) = 5 + 0.25 \cdot 3 = 5 + 0.75 = 5.75.$$

Question 36 *(1 point)* Q-learning is off-policy because its update target uses

$$r + \gamma \max_{a'} Q(x', a')$$

even when the action taken in x' was chosen by another policy.

■ True B False

Solution:
True. Q-learning uses the target

$$r + \gamma \max_{a'} Q(x', a'),$$

which depends on the greedy policy (max over a') rather than the behavior policy that actually generated the action in x'. This makes Q-learning an off-policy method.

5.4 Policy Gradient Theorem for LQR (4 points)

Consider a discounted infinite-horizon Linear Quadratic Regulator (LQR) system

$$x_{t+1} = Ax_t + Ba_t, \qquad r(x_t, a_t) = -x_t^\top C x_t - a_t^\top D a_t,$$

with discount factor $0 < \gamma < 1$, where the reward is the negative quadratic cost.
We use a linear-Gaussian policy

$$a_t \sim \mathcal{N}(Kx_t,\ \Sigma),$$

where K is the trainable policy parameter and $\Sigma \succ 0$ is fixed. Let $J(K)$ denote the expected infinite-horizon discounted reward, and let d_K denote the (normalized) discounted state distribution under the policy parameter K and $\pi_K(a_t|x_t)$ the policy under K.

Question 37 *(1 point)* The policy gradient theorem implies

$$\nabla_K J(K) = \mathbb{E}_{x\sim d_K,\ a\sim\pi_K(\cdot|x)}\Big[\left(\nabla_K \log \pi_K(a|x)\right)\ Q^{\pi_K}(x,a)\Big].$$

■ True B False

Solution:
True. By the stochastic policy gradient theorem for a discounted infinite-horizon MDP with differentiable policy $\pi_\theta(a \mid x)$, the gradient of the expected discounted return satisfies

$$\nabla_\theta J(\theta) = \mathbb{E}_{x\sim d_{\pi_\theta},\, a\sim\pi_\theta(\cdot|x)}\Big[\,(\nabla_\theta \log \pi_\theta(a \mid x))\;\; Q^{\pi_\theta}(x,a)\Big],$$

where d_{π_θ} is the discounted state distribution under policy π_θ.
In our case, the parameter is $\theta = K$ and the policy is linear-Gaussian,

$$\pi_K(a \mid x) = \mathcal{N}(Kx, \Sigma),$$

which is differentiable with respect to K. Therefore, the policy gradient theorem applies directly and yields

$$\nabla_K J(K) = \mathbb{E}_{x\sim d_K,\, a\sim\pi_K(\cdot|x)}\Big[\,(\nabla_K \log \pi_K(a \mid x))\;\; Q^{\pi_K}(x,a)\Big],$$

which is exactly the expression stated in the question.

Question 38 *(3 points)* For the linear-Gaussian policy

$$a = Kx + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \Sigma),$$

which expression for $\nabla_K \log \pi_K(a|x)$ is correct?

A $\nabla_K \log \pi_K(a|x) = (a - Kx)x^\top$

B $\nabla_K \log \pi_K(a|x) = -(a - Kx)x^\top$

■ $\nabla_K \log \pi_K(a|x) = \Sigma^{-1}(a - Kx)x^\top$

D $\nabla_K \log \pi_K(a|x) = -\Sigma^{-1}(a - Kx)x^\top$

Solution:
C. For fixed x, the policy is Gaussian

$$\pi_K(a \mid x) = \mathcal{N}(Kx, \Sigma),$$

so up to an additive constant (independent of K),

$$\log \pi_K(a \mid x) = -\frac{1}{2}(a - Kx)^\top \Sigma^{-1}(a - Kx) + \text{const.}$$

Let

$$f(K) := (a - Kx)^\top \Sigma^{-1}(a - Kx).$$

Using matrix calculus (or checking in the scalar case and generalizing), we obtain

$$\nabla_K f(K) = -2\,\Sigma^{-1}(a - Kx)x^\top.$$

Hence

$$\nabla_K \log \pi_K(a \mid x) = -\frac{1}{2}\nabla_K f(K) = -\frac{1}{2}\big(-2\,\Sigma^{-1}(a - Kx)x^\top\big) = \Sigma^{-1}(a - Kx)x^\top.$$

Therefore, the correct expression is

$$\nabla_K \log \pi_K(a \mid x) = \Sigma^{-1}(a - Kx)x^\top.$$